

Ann Mathew

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OVERVIEW

Software engineer with interdisciplinary full-stack, data analysis, and machine learning experience. Built a 4M+ Amazon seller database with Python and MySQL and analyzed 37,000+ patient medication records to improve documentation accuracy. Brings strong Python, SQL, PyTorch, TensorFlow, and data analysis skills, with a focus on data-driven products and user impact.

EDUCATION

Vanderbilt University | School of Engineering

Nashville, TN

Bachelor of Science; Majors in Computer Science, Climate & Environmental Studies

August 2022 – June 2026

- [Portfolio Site](#): built-by-ann.github.io | [GitHub](https://github.com/built-by-ann): github.com/built-by-ann
- **Relevant Coursework**: Data Structures, Algorithms, Intermediate Software Design, AI, Deep Learning, Probability & Statistics for Engineers, Database Management Systems, Senior Projects in Web Systems, Network Analysis in Healthcare, Projects in UX/UI
- **Study Abroad**: Siena, Italy (Spring 2025)

SKILLS

- **Languages**: Python, JavaScript/TypeScript, SQL, Java, C/C++, HTML/CSS
- **Frameworks & Technologies**: OOP, React Native, Node.js, REST APIs, API Design, LLM Integration, Tailwind CSS
- **Developer Tools**: Git/GitHub, Docker, Agile, AI-Assisted Dev (Claude Code/Copilot), Unit Testing, End-to-End Testing
- **Data & ML Tools**: Pandas, NumPy, scikit-learn, TensorFlow, PyTorch, GeoPandas, Tableau, Power BI, QGIS

PROFESSIONAL EXPERIENCE

Vanderbilt University Medical Center | Department of Biomedical Informatics

Nashville, TN

Research Intern [SQL, Data Analysis, Clinical Informatics]

May 2024 – August 2024

- Collaborated with Dr. Allison McCoy (PhD) to assess gaps in EHR medication records, contributing to recommendations for improving the accuracy of medication documentation in electronic health records, specifically when medications are no longer being taken.
- Conducted a comprehensive analysis of medication discontinuation practices through evaluation of 37,000+ patient encounters and interviewing physicians across various specialties, uncovering key inconsistencies and inaccuracies in medication documentation in the EHR.

TrackSellers | Amazon Marketplace E-Commerce Analytics Startup

Remote

Data Science Intern [Python, MySQL, Web Scraping, ETL Pipelines]

May 2022 – August 2022

- Built a database of 4M+ Amazon sellers by web-scraping with Python (Scrapy, XPath) and storing the data in MySQL, producing detailed seller profiles and rankings that improved partnership targeting within the Amazon ecosystem
- Facilitated connections between suppliers and sellers through tailored ranking reports and insights on competition and market niches.

PROJECTS

MedLens (medlenshealth.com) | AI-Powered Medication Reconciliation Platform

Nashville, TN

Independent Software Engineer [React, TypeScript, FastAPI, PostgreSQL, Docker, LLM Integration, AWS]

June 2026 – Present

- Engineered a full-stack clinical documentation reconciliation platform that ingests clinical notes, extracts structured medication information using AI, and automatically detects discrepancies between documentation and medication records for providers to efficiently review and reconcile.
- Designed and implemented a hybrid reconciliation engine that combines LLM-based extraction with deterministic validation, producing evidence-backed results while remaining explainable, testable, and provider-agnostic.

MorningCall | AI-Powered Full-Stack Mobile Application

Nashville, TN

Full-Stack Developer for Projects in Web Systems [Full-Stack Development, React Native, REST-APIs]

January 2026 – May 2026

- Developed end-to-end features for an AI-powered mobile productivity assistant, implementing personalized morning routines, habit tracking, task management, and conversational AI within a cohesive user experience.
- Engineered scalable frontend and backend functionality that connected mobile interfaces, secure APIs, authentication, and persistent user data to deliver personalized productivity workflows.

Land Conservation Risk Modeling | Geospatial Environmental Machine Learning

Nashville, TN

Machine Learning Project Developer for Vanderbilt Data Science [Machine Learning, Python, Data Pipelines]

November 2023 – May 2025

- Developed machine learning models using Landsat satellite imagery and geospatial environmental data to identify high-risk land areas for conservation in collaboration with the Land Trust for Tennessee.
- Produced geospatial raster datasets, methodology documentation, and an interactive web application for environmental data visualization and conservation-focused decision support.